1 Sets & Their Representation

A **set** is a well-defined collection of *distinct* objects, called its **elements**. “Well-defined” means membership is unambiguous. **Order and repetition do not matter:** $\{1,2,2,3\} = \{1,2,3\} = \{3,2,1\}$.

- **Roster (tabular) form:** list the elements, e.g. $A = \{2,4,6,8,10\}$.
- **Set-builder form:** state the defining rule, e.g. $A = \{\, x : x = 2n,\ n \in \mathbb{N},\ n \le 5 \,\}$.
- **Membership:** $x \in A$ (“belongs to”), $x \notin A$ (“does not belong to”).

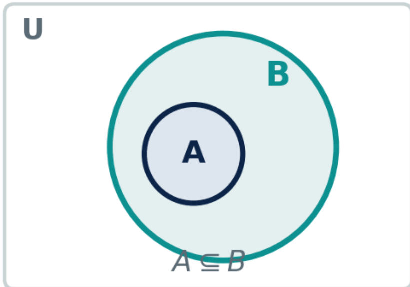
Standard number sets

- $\mathbb{N} = \{1,2,3,\dots\}$ (naturals); $\mathbb{W} = \{0,1,2,3,\dots\}$ (whole).
- $\mathbb{Z} = \{\dots,-2,-1,0,1,2,\dots\}$ (integers); $\mathbb{Q} = \left\{\dfrac{p}{q} : p,q \in \mathbb{Z},\ q \ne 0\right\}$ (rationals).
- $\mathbb{R}$ = all reals; $\mathbb{C} = \{\, a+bi : a,b \in \mathbb{R} \,\}$ (complex).
- **Chain:** $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$.

Remember: a set has no duplicate elements and no fixed order. The elements themselves can be sets (e.g. $P(A)$). “$\subseteq$” relates *sets*; “$\in$” relates an *element* to a set — don’t mix them up.

2 Types of Sets

- **Empty / null:** $\varnothing = \{\,\}$, $n(\varnothing) = 0$. **Singleton:** one element, $n(A) = 1$.
- **Finite / infinite:** a countable vs. uncountable number of elements.
- **Equal:** $A = B \iff A \subseteq B$ and $B \subseteq A$ (identical elements).
- **Equivalent:** $A \leftrightarrow B \iff n(A) = n(B)$ (same cardinality).
- **Subset:** $A \subseteq B \iff (x \in A \Rightarrow x \in B)$.
- **Proper subset:** $A \subset B \iff A \subseteq B$ and $A \ne B$.
- **Universal set** U: contains everything under discussion.



Every element of A lies in B.

Key facts: $\varnothing \subseteq A$ for every set A; $A \subseteq A$; if $A \subseteq B$ and $B \subseteq A$ then $A = B$. The empty set and A itself are the two *improper* subsets of A.

3 Subsets & the Power Set

The **power set** $P(A) = \{\, S : S \subseteq A \,\}$ is the set of *all* subsets of A (including $\varnothing$ and A).

Example. $A = \{a,b,c\}$ has $2^3 = 8$ subsets: $\varnothing,\ \{a\},\{b\},\{c\},\ \{a,b\},\{b,c\},\{c,a\},\ \{a,b,c\}$. Proper subsets: $8-1 = 7$.

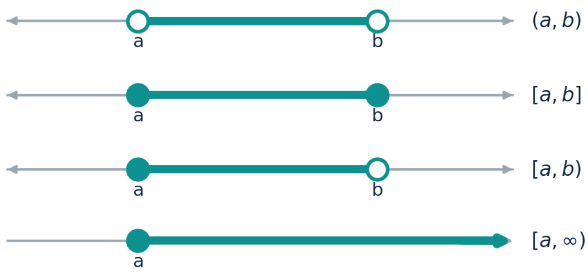
For a set with $n(A) = n$ elements:

Quantity	Formula
Number of subsets	2^n
Number of proper subsets	$2^n - 1$
Number of non-empty subsets	$2^n - 1$
Number of non-empty proper subsets	$2^n - 2$
Cardinality of the power set $n(P(A))$	2^n
Power set of a power set $n(P(P(A)))$	2^{2^n}
Subsets with exactly r elements	$\binom{n}{r}$
Subsets containing a fixed element	2^{n-1}

4 Intervals on the Real Line

Continuous subsets of $\mathbb{R}$ are written as intervals.

Interval	Set-builder
(a, b)	$\{x : a < x < b\}$ (open)
$[a, b]$	$\{x : a \le x \le b\}$ (closed)
$[a, b)$	$\{x : a \le x < b\}$
$(a, b]$	$\{x : a < x \le b\}$
$[a, \infty)$	$\{x : x \ge a\}$
$(-\infty, b)$	$\{x : x < b\}$
$(-\infty, \infty)$	$\mathbb{R}$

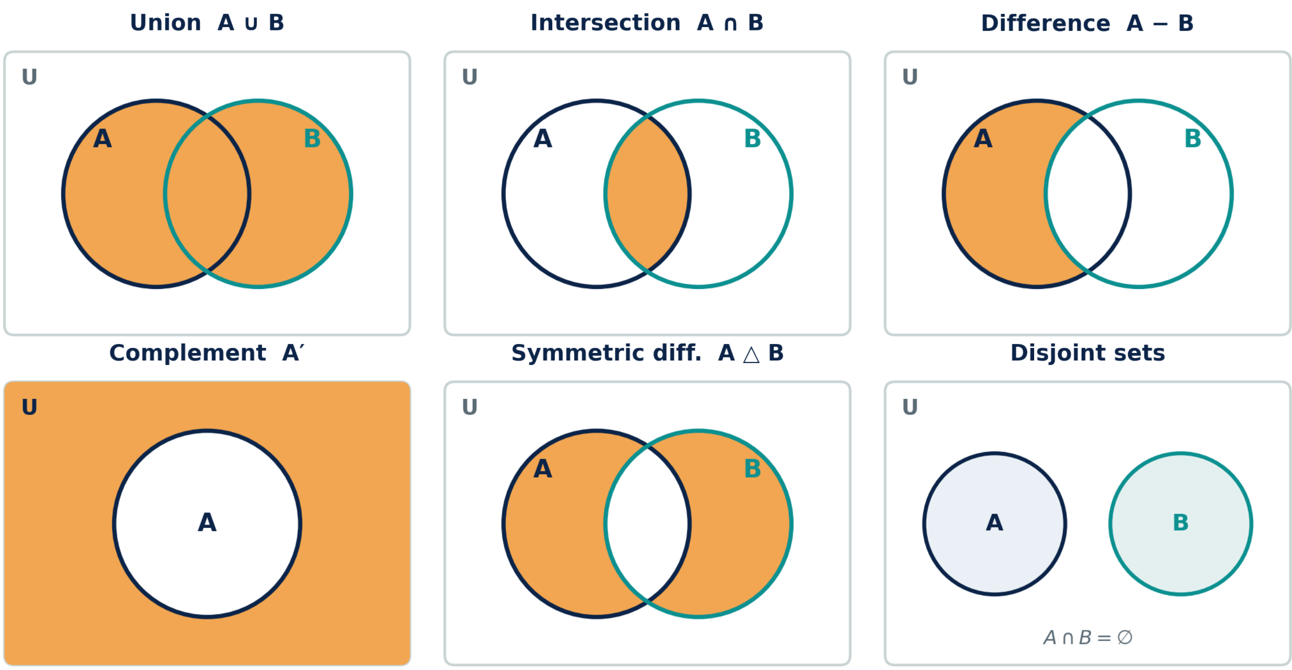


Filled dot = included, open dot = excluded.

- Length of a bounded interval $= b - a$. Endpoints with ∞ are always *open*.

5 Set Operations

Operation	Definition
Union $A \cup B$	$\{\, x : x \in A$ or $x \in B \,\}$
Intersection $A \cap B$	$\{\, x : x \in A$ and $x \in B \,\}$
Difference $A - B$	$\{\, x : x \in A$ and $x \notin B \,\}$
Complement A'	$U - A = \{\, x \in U : x \notin A \,\}$
Symmetric difference $A \triangle B$	$(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$
Disjoint sets	$A \cap B = \varnothing$ (no common element)



Example. $U = \{1, \ldots, 9\},\ A = \{1, 2, 3, 4\},\ B = \{3, 4, 5, 6\}$: $\quad A \cup B = \{1, 2, 3, 4, 5, 6\}, \quad A \cap B = \{3, 4\}, \quad A - B = \{1, 2\}, \quad B - A = \{5, 6\}, \quad A' = \{5, 6, 7, 8, 9\}, \quad A \triangle B = \{1, 2, 5, 6\}.$

6 Algebra of Sets — Laws

- **Idempotent:** $A \cup A = A, \quad A \cap A = A.$
- **Identity:** $A \cup \varnothing = A,\ \ A \cap U = A; \qquad A \cup U = U,\ \ A \cap \varnothing = \varnothing.$
- **Commutative:** $A \cup B = B \cup A, \quad A \cap B = B \cap A.$
- **Associative:** $(A \cup B) \cup C = A \cup (B \cup C), \quad (A \cap B) \cap C = A \cap (B \cap C).$
- **Distributive:** $A \cup (B \cap C) = (A \cup B) \cap (A \cup C), \quad A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$
- **Complement:** $A \cup A' = U,\ \ A \cap A' = \varnothing; \qquad (A')' = A,\ \ U' = \varnothing,\ \ \varnothing' = U.$
- **Absorption:** $A \cup (A \cap B) = A,\ \ A \cap (A \cup B) = A; \qquad A \cup (A' \cap B) = A \cup B.$

Duality: swap $\cup \leftrightarrow \cap$ and $U \leftrightarrow \varnothing$ in any true identity and you get another true identity.

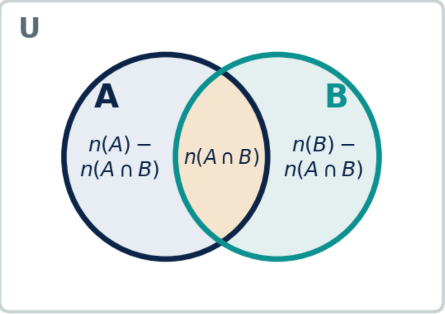
7 De Morgan's Laws

- $(A \cup B)' = A' \cap B', \qquad (A \cap B)' = A' \cup B'.$
- **Generalized:** $\left(\bigcup_i A_i\right)' = \bigcap_i A_i', \qquad \left(\bigcap_i A_i\right)' = \bigcup_i A_i'.$
- **Difference form:** $A - (B \cup C) = (A - B) \cap (A - C), \qquad A - (B \cap C) = (A - B) \cup (A - C).$

Example. $U = \{1, \ldots, 6\},\ A = \{1, 2, 3\},\ B = \{3, 4\}$: $(A \cup B)' = \{5, 6\}$ and $A' \cap B' = \{4, 5, 6\} \cap \{1, 2, 5, 6\} = \{5, 6\}$. ✓

8 Venn Diagrams & Cardinality (Counting)

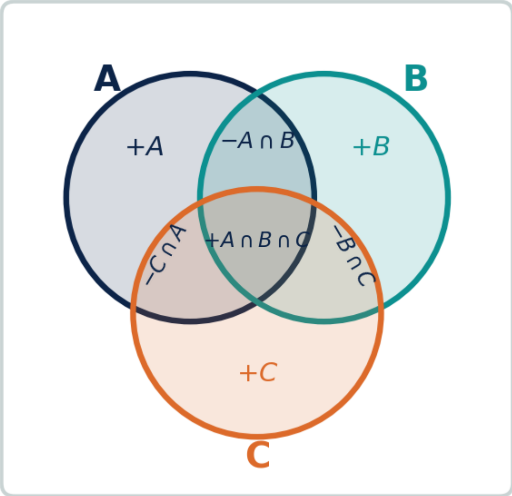
- **Two sets:** $n(A \cup B) = n(A) + n(B) - n(A \cap B)$; disjoint $\Rightarrow n(A \cup B) = n(A) + n(B)$.
- **Partition:** $n(A \cup B) = n(A - B) + n(A \cap B) + n(B - A).$
- $n(A - B) = n(A) - n(A \cap B) = n(A \cap B')$; $\ n(A') = n(U) - n(A).$
- **Neither:** $n\big((A \cup B)'\big) = n(U) - n(A \cup B).$
- $n(A \triangle B) = n(A) + n(B) - 2\,n(A \cap B).$



Each region is counted once.

Example. In a class of 40: 25 play cricket, 20 play football, 10 play both. At least one $= 25 + 20 - 10 = 35$; neither $= 40 - 35 = 5$; only cricket $= 15$; only football $= 10$.

Three sets & selection counts



+ singles − pairs + triple.

- $n(A \cup B \cup C) = n(A) + n(B) + n(C)$
  $-n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$.
- **General** n**:** $n\left(\bigcup_{i=1}^{n} A_i\right) = \sum_i n(A_i) - \sum_{i<j} n(A_i \cap A_j) + \cdots + (-1)^{n+1} n\left(\bigcap_{i=1}^{n} A_i\right)$.
- **Exactly one** of A, B: $n(A) + n(B) - 2n(A \cap B)$.
- **Exactly one** of A, B, C: $\sum n(A) - 2\sum n(A \cap B) + 3n(A \cap B \cap C)$.
- **Exactly two:** $\sum n(A \cap B) - 3n(A \cap B \cap C)$.
- **At least two:** $\sum n(A \cap B) - 2n(A \cap B \cap C)$.
- **None:** $n(U) - n(A \cup B \cup C)$.

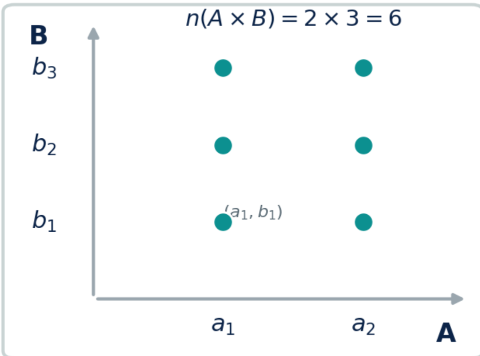
Example. Group of 100: 40 read A, 30 read B, 30 read C; 12 read $A\&B$, 8 read $B\&C$, 10 read $A\&C$, 5 read all three. At least one $= 40 + 30 + 30 - 12 - 8 - 10 + 5 = 75$; none $= 100 - 75 = 25$.

9 Cartesian Product

The **Cartesian product** $A \times B = \{\, (a, b) : a \in A,\ b \in B \,\}$ is the set of all *ordered pairs*.

- $n(A \times B) = n(A) \cdot n(B)$; in general $A \times B \neq B \times A$.
- **Distributive:** $A \times (B \cup C) = (A \times B) \cup (A \times C)$;
  $A \times (B \cap C) = (A \times B) \cap (A \times C)$;
  $A \times (B - C) = (A \times B) - (A \times C)$.
- Number of relations from A to B: 2^{pq}, $p = n(A)$, $q = n(B)$.

Example. $A = \{1, 2\}$, $B = \{x, y\}$: $A \times B = \{(1, x), (1, y), (2, x), (2, y)\}$, $n = 2 \times 2 = 4$.



$2 \times 3 = 6$ ordered pairs.

10 Symmetric Difference — Properties

- **Commutative:** $A \triangle B = B \triangle A$. **Associative:** $(A \triangle B) \triangle C = A \triangle (B \triangle C)$.
- $A \triangle A = \varnothing$, $A \triangle \varnothing = A$, $A \triangle U = A'$.
- $A \triangle B = \varnothing \iff A = B$; $A \triangle B = (A \cup B) - (A \cap B)$.

Example. $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$: $A \triangle B = (A - B) \cup (B - A) = \{1, 2\} \cup \{5, 6\} = \{1, 2, 5, 6\}$.

11 Must-Know Identities

- **Subset chain:** $A \subseteq B \iff A \cap B = A \iff A \cup B = B \iff B' \subseteq A'$.
- $A - B = A \cap B'$; $A - (A \cap B) = A - B$.
- $(A - B) - C = A - (B \cup C)$; $A - (B - C) = (A - B) \cup (A \cap C)$.
- **Absorption:** $A \cup (A \cap B) = A$, $A \cap (A \cup B) = A$.

12 Quick Recall & Common Mistakes

Remember

- n-element set: 2^n subsets, $2^n - 1$ proper; $n(P(A)) = 2^n$.
- Draw the Venn diagram first — counts read straight off it.
- $A \subseteq B \iff A \cap B = A \iff A \cup B = B$.
- De Morgan flips $\cup \leftrightarrow \cap$ under complement.
- Inclusion–Exclusion: $+$ singles $-$ pairs $+$ triple.

Avoid these slips

- $\varnothing \neq \{\varnothing\}$: the first is empty, the second has one element.
- $\{0\} \neq \varnothing$ and $\{\varnothing\} \neq \varnothing$.
- $A \times B \neq B \times A$ unless $A = B$ (or one is empty).
- $\in$ links element–set; $\subseteq$ links set–set.
- In Inclusion–Exclusion, remember to *add back* the triple overlap.